SS1 MATHEMATICS – THIRD TERM

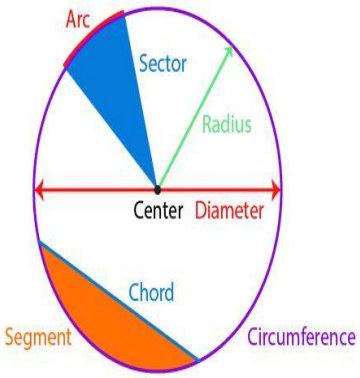
WEEK 1: MENSURATION I (CIRCLE)

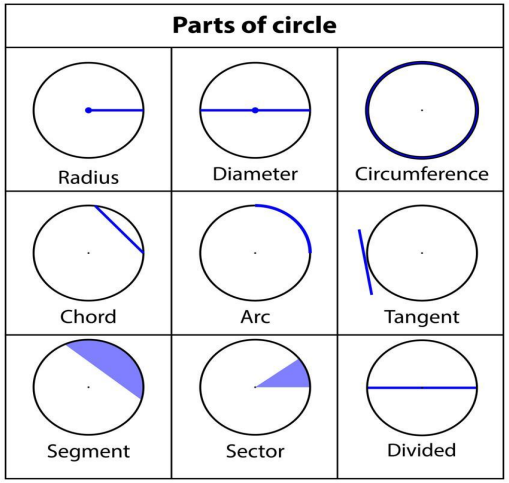
1. INTRODUCTION (Short Note)

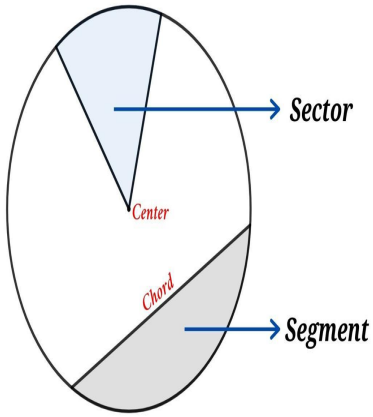
Mensuration deals with **measurement of shapes** (length, area, volume).

A **circle** is a plane shape where all points on the boundary are the same distance from a central point.

Parts of a Circle







- **Radius (r):** Distance from center to edge
- **Diameter (d):** Distance across circle through center
- **Chord:** Line joining two points on circle
- **Arc:** Part of the circumference
- **Sector:** Region between two radii and an arc
- **Segment:** Region between a chord and an arc

2. BASIC FORMULAS

- Diameter:
  $d = 2r$
- Circumference:
  $C = 2\pi r$ OR $C = \pi d$
- Area:
  $A = \pi r^2$
- Use:
  $\pi = 22/7$ OR $\pi = 3.142$

3. LENGTH OF ARC

Concept

Arc length is a **fraction of the circumference**.

Formula

Arc Length = $(\theta / 360) \times 2\pi r$

Where:

- θ = angle at center (in degrees)
- r = radius

Example 1

Find arc length when:
$r = 21$ cm, $\theta = 60°$, $\pi = 22/7$

Arc Length = $(60 / 360) \times 2 \times 22/7 \times 21$
$= (1 / 6) \times 132$
= **22 cm**

4. FINDING ANGLE FROM ARC

Formula

Arc Length = $(\theta / 360) \times 2\pi r$

Make θ the subject:

θ = (Arc Length × 360) / ($2\pi r$)

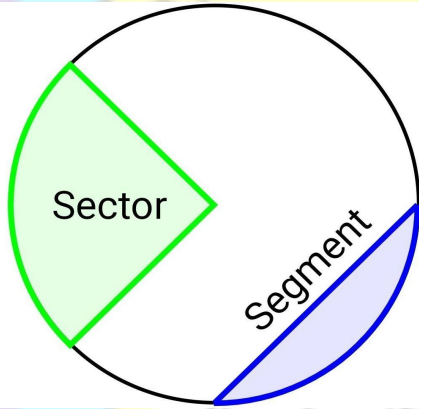
Example 2

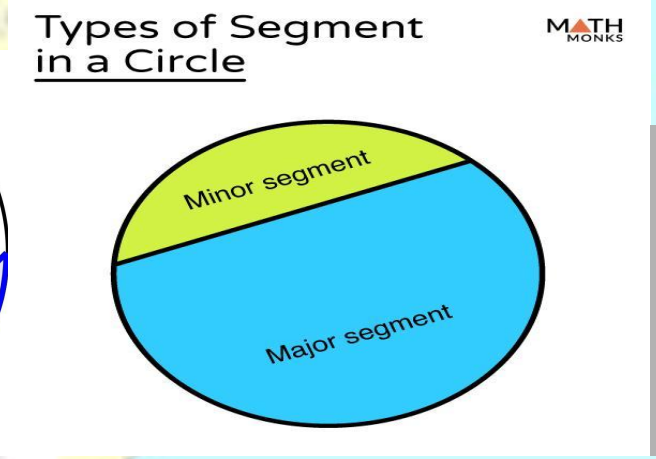
Arc = 11 cm, $r = 7$ cm

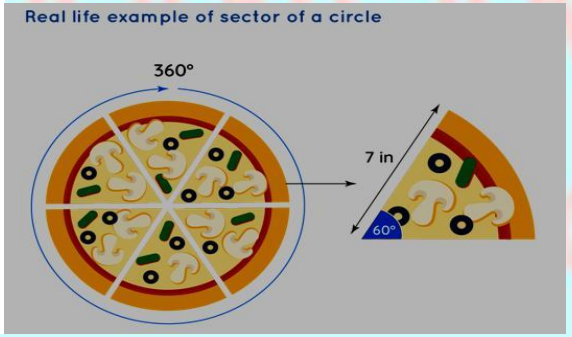
$\theta = (11 \times 360) / (2 \times 22/7 \times 7)$
$= 3960 / 44$
= **90°**

5. SECTORS AND SEGMENT

Visual Understanding







6

- **Sector:** Like a pizza slice
- **Segment:** Between chord and arc

6. PERIMETER OF A SECTOR

Concept

Perimeter = Arc length + 2 radii

Formula (WPS Format)

Perimeter = (θ / 360) × 2πr + 2r

Example 3

r = 14 cm, θ = 120°

Arc = (120 / 360) × 2 × 22/7 × 14
= (1 / 3) × 88
= 88/3

Perimeter = 88/3 + 28
= (88 + 84) / 3
= **172/3 cm ≈ 57.33 cm**

7. KEY POINTS TO REMEMBER

- Full circle = **360°**
- Arc is **part of circumference**
- Sector = **area slice**
- Segment = **cut by chord**
- Always convert angle into fraction of 360

8. REVISION QUESTIONS

1. State the formula for circumference
2. What is a chord?

3. Differentiate between sector and segment
4. What makes up the perimeter of a sector?
5. How many degrees in a circle?

9. EVALUATION (CLASSWORK)

1. Define a sector
2. Define a segment
3. Write arc length formula
4. Find arc length when $r = 10$ cm, $\theta = 90°$ (leave in π)
5. Find perimeter if $r = 5$ cm and arc = 10 cm

10. ASSIGNMENT

1. Find arc length:
   $r = 28$ cm, $\theta = 45°$
2. Perimeter = 25 cm, $r = 7$ cm
   (a) Find arc length
   (b) Find θ
3. Wheel radius = 35 cm, 100 turns
   Find distance covered
4. Clock minute hand = 10 cm
   Distance moved in 20 minutes

CANAANLAND
MODEL GROUP OF SCHOOLS
GOD IS OUR STRENGTH